1.1 Introduction

Project Name:	Fault Detection and Energy Monitoring System with Adaptive Analysis
Organization:	Ahsanullah University of Science and Technology
Location:	Dhaka, Bangladesh
Duration:	6 months
Position:	Team Leader

1.2 Background

A. Overview

A fault detection and energy monitoring system is a sophisticated electrical system solution meant to improve safety, reliability, and efficiency in contemporary power systems. It operates on a watch to check the electrical parameters of voltage, current, and power to detect any abnormal situations, which could be earth faults and leakage currents. It combines real-time monitoring, data logging and analytical functions as opposed to standard protection devices. The system applies adaptive detection tools, signal, and IoT-based communication to enhance precision and responsiveness. It helps in making suitable decisions, anticipatory repair, and effective use of energy at homes, factories, and laboratory settings.

The project was initiated to come up with an elaborate electrical monitoring system that deals with the shortcomings of traditional fault protection devices. It aimed at increasing the safety, reliability and energy efficiency via ongoing observation and smart analysis. A modular system structure was adopted, and the system comprised multi-point voltage sensing in Phase-Neutral, Phase-Earth, and Earth-Neutral paths, and current measurement. An external ADC gave the data high resolution, with the signal conditioning methods of voltage scaling, filtering, and offset-correction. Electrical parameters, such as the voltage, current, and power, were calculated to facilitate the analysis of the system. An adaptive fault detection mechanism using statistical thresholding was proposed to enhance the reliability of the detection with the operation under different operating conditions. The project also included real-time visualization, display units and remote monitoring using the IoT. Simulation, signal processing and analytical assessment were simulated using MATLAB-based tools to create a complete workflow of intelligent electrical systems management.

B. Objective

The primary purpose was to come up with a smart embedded system of real-time electrical monitoring and adaptive fault detection to ensure better safety, accurate measurement, energy analysis, and steady system performance across different operating environments.

The specific objectives were:

- To acquire the parameters of voltage and current with high-resolution sensing and signal conditioning instruments.
- To achieve adaptive fault detection with statistical thresholding to achieve greater reliability and less false triggering.
- To allow real-time tracking and remote data recording with the help of IoT communication and analysis.

C. Nature of Work

I did the design and development of a smart electrical monitoring system aimed at fault detection and energy analysis. I analyzed system requirements and chose suitable sensors, a microcontroller and communication modules. I was using data acquisition in such a way that I connected the voltage and current sensors to a high-resolution ADC. I created signal conditioning methods such as voltage scaling, filtering and offset correction to enhance the accuracy of the measurements. I created the procedures for parameter computation of voltage, current and power. I developed an adaptive fault detection algorithm through statistical thresholding to make it more reliable. I incorporated IoT-based communication for real-time monitoring and data logging on the cloud. I also performed MATLAB analysis to validate, graphically chart, and estimate errors. I put the system through normal and faulting tests to measure performance, which was accurate, responsive and system reliability.

D. Hierarchical Structure

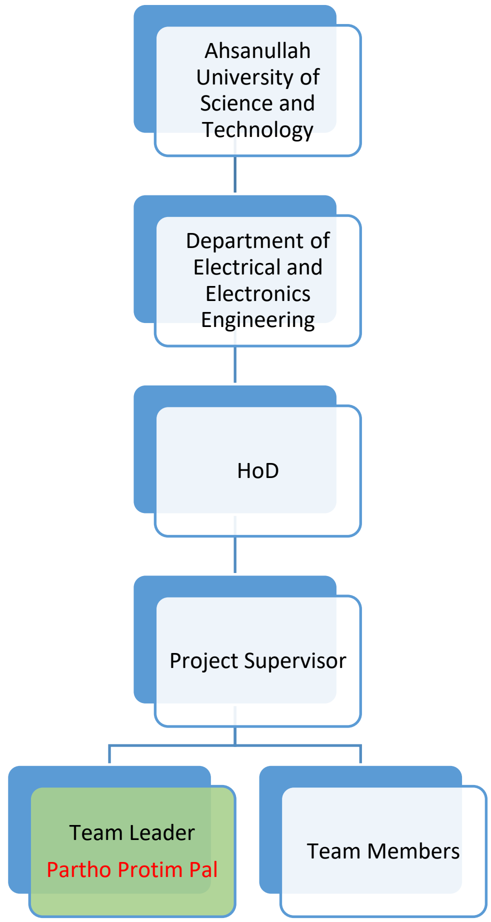

Figure 1: Organizational Chart

E. Duties and Responsibilities

- To study the drawbacks of electrical systems and set the requirements of sophisticated fault detection and energy monitoring.
- To choose and combine appropriate sensors, microcontroller and communication modules to obtain and process data correctly.
- To develop signal conditioning and parameter computation techniques for reliable electrical measurements.
- To create an adaptive fault detection algorithm and combine monitoring and MATLAB analysis using IoT.
- To test system performance by analyzing data and testing it under normal and faulty conditions.

1.3 Personal Engineering Activities (PEA)

A.

I started the project with the recognition of the increasing necessity of constant monitoring, operational safety and energy saving in contemporary electrical systems. I also understood that earth faults were major dangers due to leakage currents to ground, such as equipment damage, fire and safety issues. I critically evaluated traditional protection equipment, including ELCBs and RCDs and ascertained that though valid in disconnection under the fault, they did not contain real-time monitoring, data capture, and analytical faculties. By this gap, I established the project objective as developing an intelligent embedded system that could do improved fault detection and energy monitoring. I developed a conceptualized system architecture that includes multi-point voltage sensing over Phase-Neutral, Phase-Earth and Earth-Neutral paths and current sensing. I added adaptive fault detection logic in order to enhance reliability when the conditions are dynamic. I also combined IoT-driven communications and MATLAB data analytics to guarantee advanced monitoring and visualization, as well as decision-making driven by data to enhance the control of electrical systems.

B.

To get the right measurement and proper fault detection, I selected the right sensing, processing and communication components to come up with the right system parameters. I chose the ESP32 microcontroller as a central processing unit, because of the 32-bit architecture, built-in Wi-Fi, and a variety of communication ports, it to effectively process data and communicate with the IoT. In my design, I used the ADS1115 16-bit ADC to achieve high-resolution signal acquisition because it was more accurate than internal converters. I put three ZMPT101B voltage sensors that would measure Phase-Neutral, Phase-Earth, and Earth-Neutral voltages in place to ensure a full range of fault coverage. In the case of the current measurement, I used an ACS712 Hall-effect sensor to precisely measure load current. I added an LCD that enables the real-time visualization of the system and a buzzer, and an LED that serves as a fault conditions alert. Some of the supports that I also used included voltage division resistors, noise filter capacitors, and a regulated 5V power supply, which would maintain system stability as well as reliability of measurements.

C.

I created the original methodology on a modular system design with steps of sensing, signal conditioning, and computation of parameters. I used data acquisition by recording voltage signals in Phase-Neutral and Phase-Earth and Earth-Neutral paths and recording currents in a Hall-effect sensor. I connected all analog signals to the ADS1115 ADC so that digital conversion would be high-resolution. I created a signal conditioning process that consisted of scaling up voltages using resistive dividers, noise-cutting by use of moving average filtering, and offset correction to remove sensor bias so as to better measure temperatures. Then I calculated the electrical parameters in which the voltages were calculated by a calibrated ADC scaling and the current was calculated by sensor calibration. I obtained power in terms of the basic equation of voltage and current, so that a representation of a load behavior is accurate. The methodological approach provided a solid basis for high-quality data collection and pre-processing, which was used to achieve fault detection and system analysis.

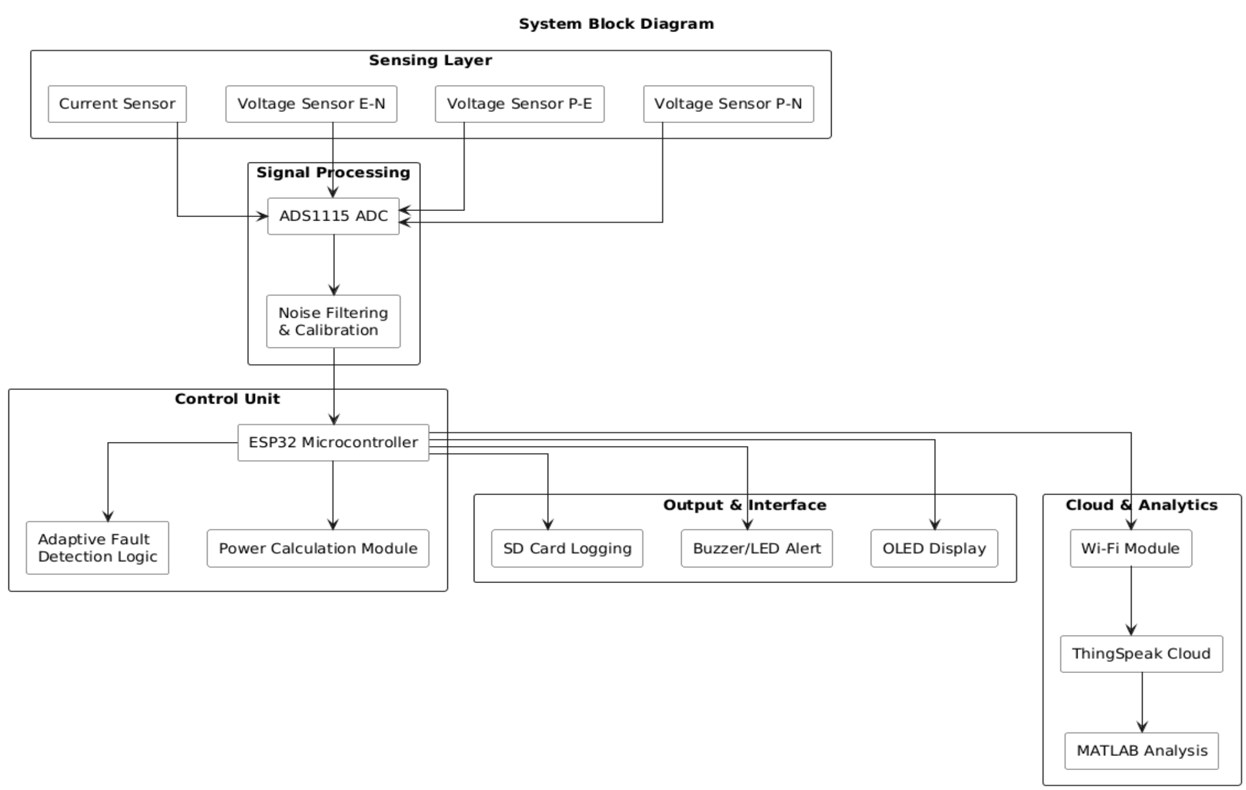

Figure 2: Rectangular Fins

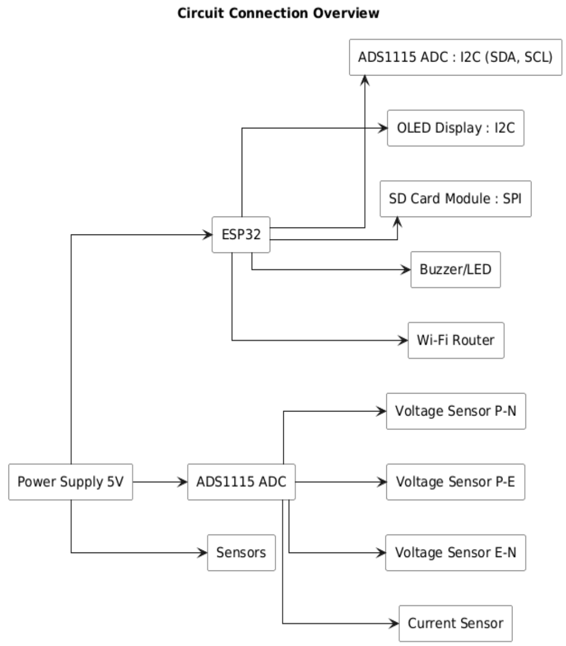

Figure 3: Circuit Connection Overview

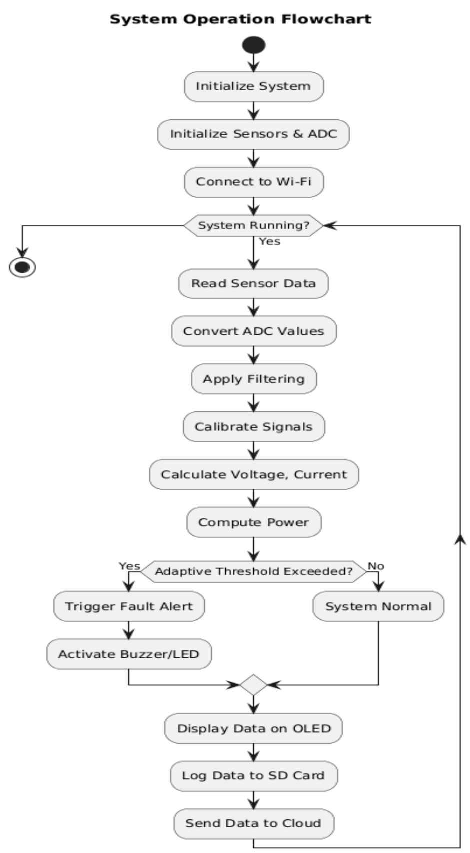

Figure 4: System Flowchart

D.

I used adaptive fault detection with the intention of countering the shortcomings of the old-fashioned fixed threshold system. I established a statistical density level using the mean and standard deviation, with a sensitivity to freely vary the detection limits, depending on different operational conditions. I also set fault conditions when any of the Phase-Earth or Earth-Neutral voltages were above the adaptive threshold to provide a greater detection accuracy and a smaller false trigger. I incorporated live data visualization on an LCD to display voltage, current, power, and fault status to have immediate knowledge of the system. I also designed an IoT network of communication with an ESP32 Wi-Fi module to allow data logging and monitoring remotely in the cloud. To test the performance of the system, I applied MATLAB to simulate, filter, and visualize faults. I performed graphical analysis of voltages, current, power trends and comparison of fault and evaluation of error. This thorough methodology enabled proper detection, good performance and efficient analytical capacity.

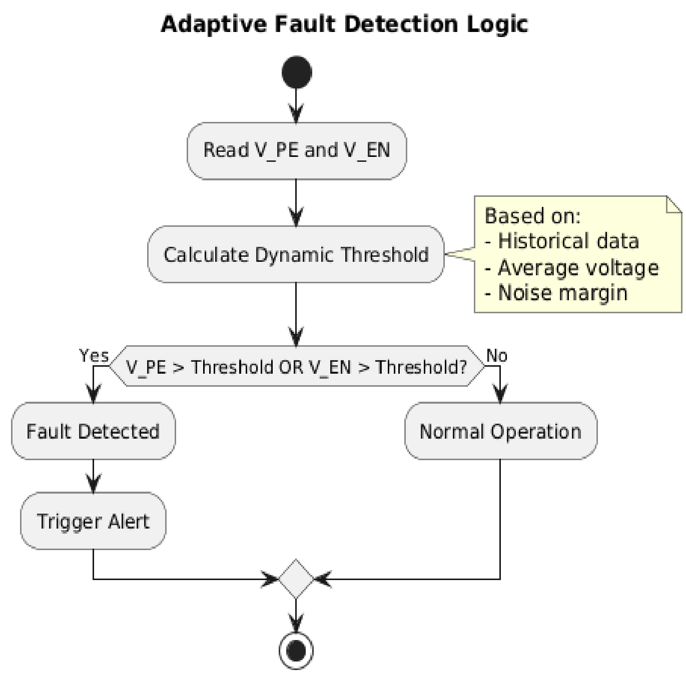

Figure 5: Adaptive Fault Detection Logic

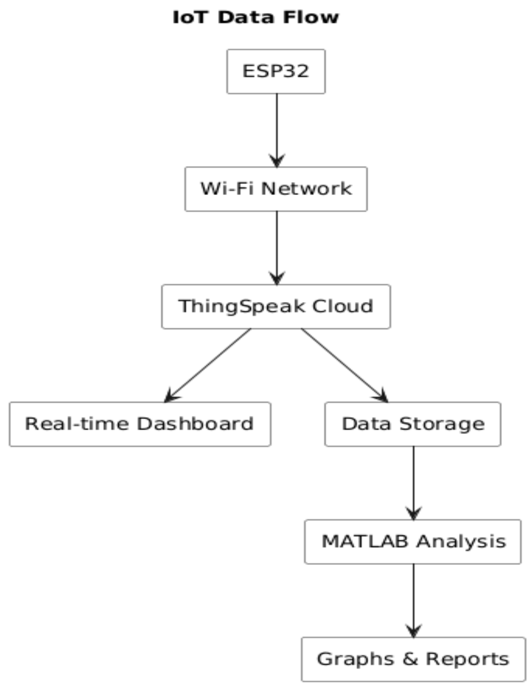

Figure 6: IoT Data Flow

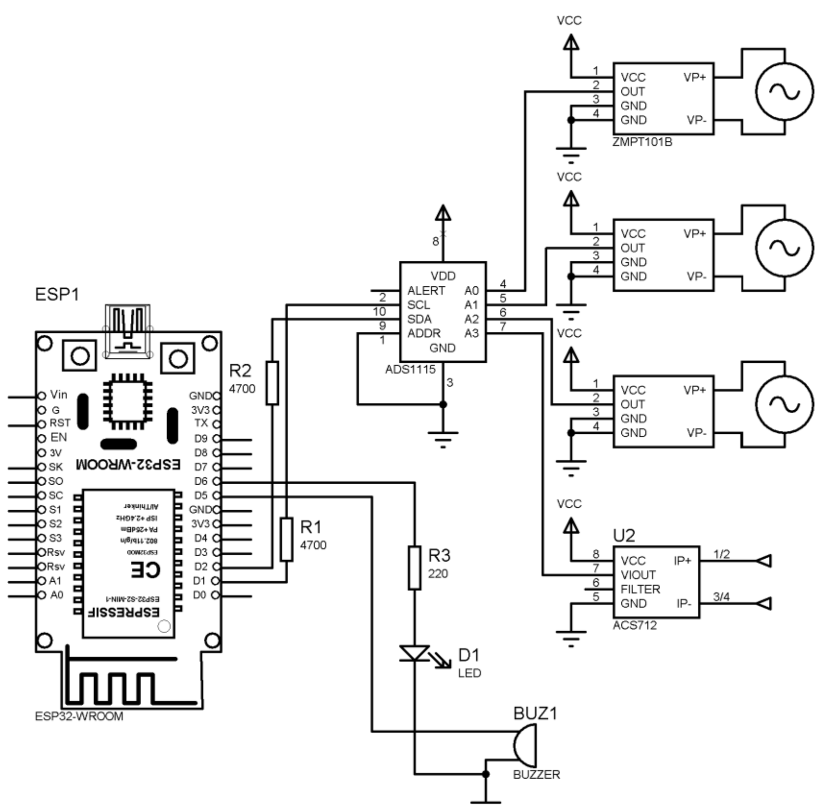

Figure 7: Circuit Design

E.

I simulated and processed data to assess the system performance at normal and fault conditions. I measured a constant Phase-Neutral voltage of approximately 230 V when at stable operation, with Phase-Earth and Earth-Neutral voltages being low, which indicated that the system was stable. In the course of fault simulation, I experienced a high rise in these voltages, which confirms fault sensitivity. My results provided current and power behavior, which was stable with slight fluctuations and realistic power calculations, which also showed good ability to monitor energy. The adaptive threshold method allowed me to attain fault detection with a 200-300 milliseconds with no false positives in the test. I confirmed the accuracy of the system by MATLAB-based graphical analysis, such as voltage, current, power trends, and fault visualization. I upheld measurement error at ±2-5% because of high-resolution ADC, calibration and filtering methods. I ensured that the system offered real-time control, adaptable identification, cloud record and a scalable and cost-efficient system, and listed limitations that included single-phase operation and reliance on Wi-Fi.

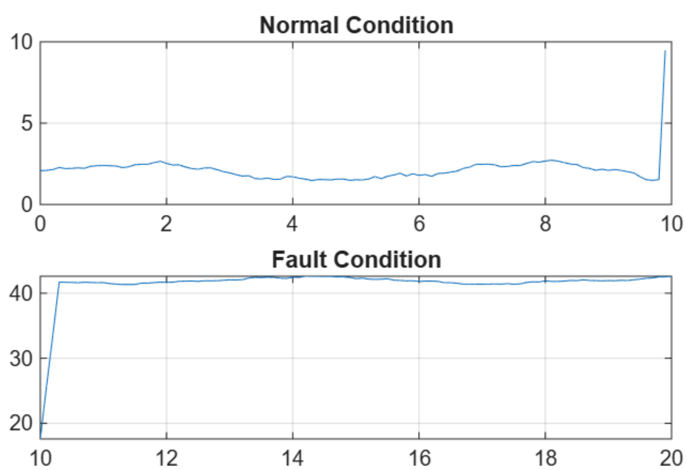

Figure 8: Comparison of Normal and Fault Condition

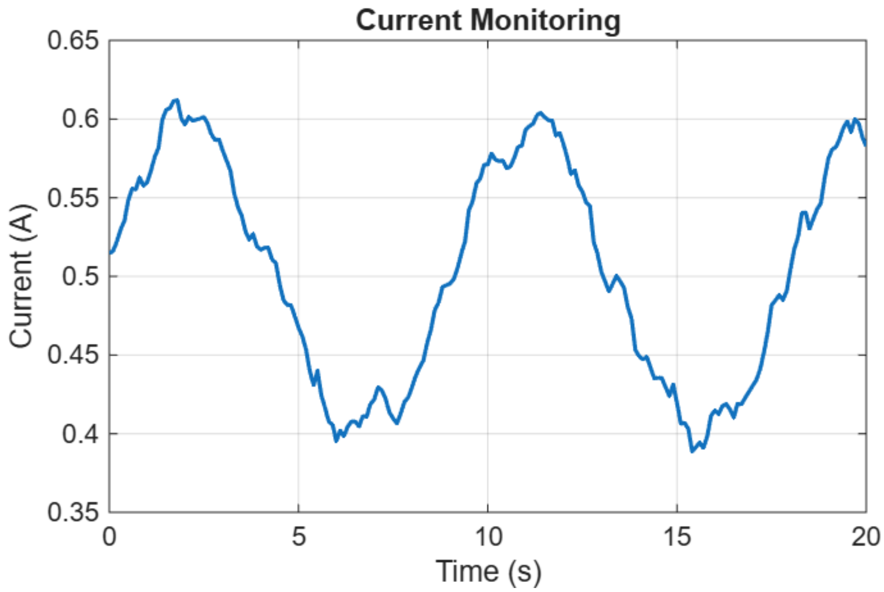

Figure 9: Current Monitoring

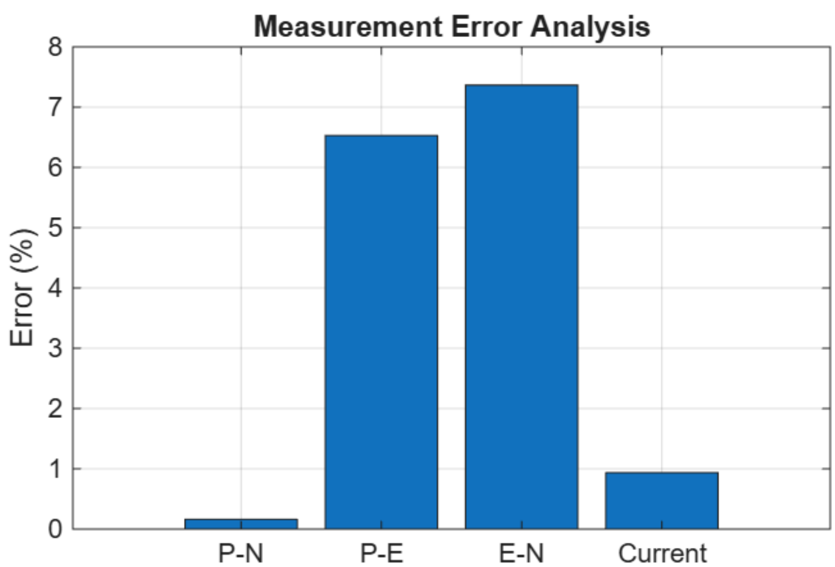

Figure 10: Measurement Error Analysis

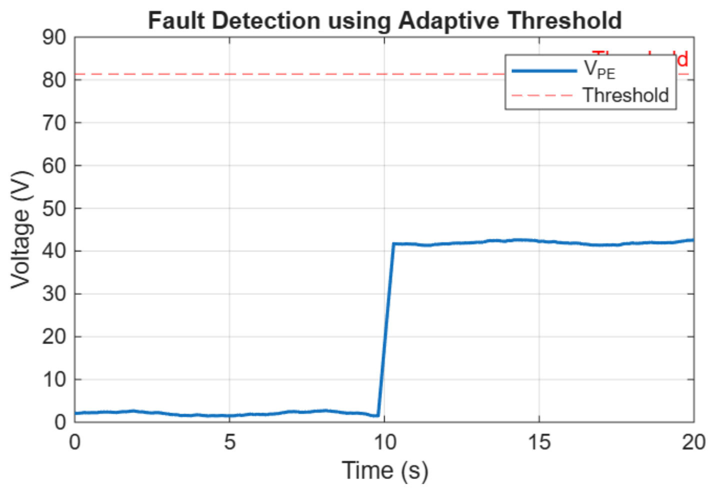

Figure 11: Fault Detection using Adaptive Threshold

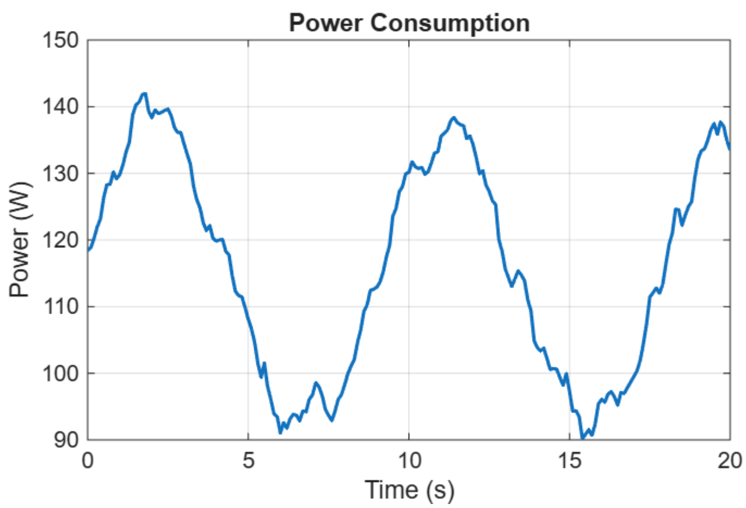

Figure 12: Power Consumption

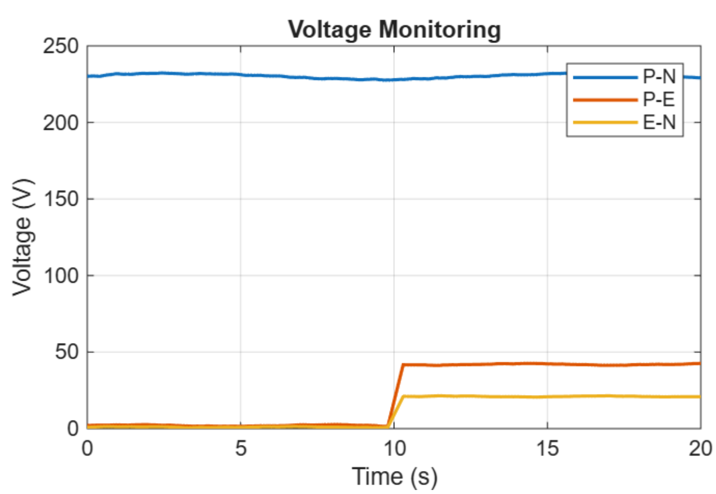

Figure 13: Voltage Monitoring

1.4 Technical Problems and Solutions

A.

I realized a significant issue in most traditional fault-detecting systems where fixed threshold-based answers, such as ELCBs and RCDs, did not deliver acceptable detection in differing operating conditions. The area of impact was the validity and acceptability of earth fault detection in dynamic electric conditions. It was caused by the fact that the threshold limits used were set at a fixed value and did not respond to changes in the voltage signals, similar to missed faulting or false triggering. This was a major limitation to system safety and operational reliability. To address this, I introduced an adaptive fault detection algorithm using statistical parameters, where the mean and standard deviation with a sensitivity factor are used as the threshold. I implemented this reason in the ESP32 processing unit. This led to successful fault detection with a response time of about 200-300 milliseconds, and false positives were removed in the system testing.

B.

I also had a technical issue regarding faulty voltage and current measurements of signal noise and sensor discrepancies during the data collection process. The parameter reliability in calculating power and the ability to detect faults were the affected areas. The basic problem was the noise intrusion in the analog signal and offset in sensor outputs, resulting in errors in measured values. This problem affected the system performance by creating possible mistakes in analysis and monitoring. To overcome this, I implemented a signal conditioning process with resistive scaling of voltage, moving average averaging of noise and offset correction of sensor calibration. I used a high-resolution ADS1115 16-bit ADC to enhance the accuracy of measurement. Consequently, I minimized measurement error within ±2-5% and achieved constant and accurate data capture to be able to have consistent system operation and fault detection.

1.5 Creative Work

I created a dynamic fault detector that employs a statistical threshold instead of a fixed threshold to enhance reliability during changing conditions. I creatively combined multi-point voltage monitoring and monitoring based on the Internet of Things with MATLAB analysis. I also generated an effective signal conditioning strategy to raise the precision of measurements, rendering a smart, scaled and low-priced electrical monitoring system.

1.6 Team Coordination

I organized the project activities by clearly stating the tasks, roles and enabled proper communication among team members in the process of development. I used workflow by breaking down the project into modules like sensing, processing and communication, which were also developed in parallel to ensure an efficient integration. I guided the technical discussion groups to solve the design issues regarding signal conditioning and fault detection logic in order to ensure that project goals were achieved. I also kept track of progress within specific timeframes and also made sure that every part performed and fitted its performance and accuracy targets. I also made sure that I did the documentation and shared data with consistency and transparency within the team. I managed to successfully finish the project on time and maintain the desired level of quality, safety concerns, and technical accuracy of work on all project phases due to the organized coordination, proactivity, and problem-solving.

1.7 Codes and Standards

In the design of the system, I used IEC 60364, which governs low-voltage electrical installations to consider appropriate grounding, fault protection, and wiring. I also used the IEEE 1459 standards to correctly measure electrical parameters and prescribed my voltage, current, and power monitoring approach to ensure high accuracy and reliability in measurements.

1.8 Summary

A.

The project aimed at designing a smart electrical monitoring system to improve the safety, reliability, and energy levels of current power systems. The necessity emerged as a result of shortcomings in traditional protection devices, which had no real-time monitoring and analysis. A modular system structure was developed, including multi-point voltage measurement over Phase-Neutral and Phase-Earth and Earth-Neutral circuits as well as current. A 16-bit ADC was used to obtain high-resolution data with the help of signal conditions such as scaling, filtering, and offset correction. Voltage, current, and power electrical values were calculated and an adaptive fault detection algorithm using statistical thresholds was applied to enhance higher accuracy under different conditions. It incorporated IoT-based remote monitoring and MATLAB analysis and validation. Performance testing proved correct measurements, speedy failure detection and assured functioning. The project ended up with an effective model scalable solution of monitoring smart electrical systems.

B.

I acquired good design skills in embedded systems, electrical parameter measurement and fault tolerance methods. I improved my skills in signal conditioning, data analysis and performance evaluation using MATLAB-based simulation. I also enhanced the problem-solving, work with systems, and IoT, as well as the technical documentation and analysis of decisions in multifaceted engineering settings.